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History of Timber Construction

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Abstract: The history of timber construction is presented from prehistoric to modern times, from the simple hut to multi-purpose halls, from trial-and-error construction to one based on engineering design and analysis, from simple beams and posts to modern spatial systems. The development of common timber construction forms and methods are briefly discussed. Wood, our only basic renewable resource, has been an important part of man’s built environment. For this to continue in the future, developments in timber engineering and related technologies should meet the needs and challenges demanded by building occupants and by society, in general.

Keywords: history, timber, wood, structures, house, building

Introduction

From the earliest known years of man's existence, trees and wood have provided him with fuel, tools, food and shelter. As his needs evolved, so did his use of wood. The history of timber construction is traced in this paper from prehistoric to modern times, from the simple hut to multi-purpose halls, from trial-and-error construction to one based on engineering design and analysis, from simple beams and posts to modern spatial systems. Some advanced structures today can usually be traced back to their generic ancestors in concept [1]. A few early building concepts even fulfilled their requirements, in some places of the world, so effectively that its design and construction have remained unchanged through hundreds of years. Recent technologies have, however, opened exciting new possibilities in timber construction.

Wood is the oldest building material capable of transferring tension and compression forces making it naturally suited as a beam element. It has a very high strength to weight ratio. Its workability and versatility – i.e., easily cut and joined using simple tools, materials and techniques, bent into desired shapes, and pulped or exploded into its constituent fibers or chipped into small pieces and then reconstituted – are well recognized. Wood possesses exceptional impact strength, effective thermal insulating

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properties, high electrical resistance when dry and effective resistance to atmospheric pollution, sea air and chemicals. The beauty and feelings of warmth and intimacy it offers are unmatched. Among the major construction materials, wood uses the least amount of energy to process and manufacture and, unlike other industrial materials, is renewable. The unique properties and ready availability of wood have made it a very important contributor to the advance of civilization and the development of society [2-4].

Prehistory and Early Civilizations

Early Shelter

Man's early culture consisted mainly of a struggle for food and defense. His shelter provided the latter. The earliest known use of wood in shelter is as beams to support the earth covered roof in pit dwellings or as posts in earth lodges and/or tent-like huts of branches and twigs. In some places during the Paleolithic and Mesolithic times, a ring of branches were dug firmly in the ground and bent over towards the center where they were interlaced and bound together [5] (Figures 1a and b). The use of large-diameter tree trunks or branches was initially avoided, if possible, because cutting them across the grain with the available tools (probably chipped or polished stone axes and/or animal teeth) was difficult. Interwoven branches and other vegetable materials then filled the continuous wall and roof frame. Houses on the Turkish-Iranian border (c. 9000 BC) had oval or circular stone bases containing hearths and supported a light superstructure of daub and wattle, or reeds, or matting [6]. The frame was typically made from branches that were lashed together with roots, leaf fibers or climbing plants. These early forms (Figures 1a to c) seem to have been strongly influenced by cave shelters in shape and plan.

Other building shapes, such as conical domes and cylindrical lattice houses with rectangular plans, appeared c. 8000-7000 BC (e.g., Figures 1d to f). Neolithic settlements continued the trend for round hut shelters but the use of rectangular ground plans became more widespread [6], probably because the use of heavier horizontal beams for walls and roofs favored a rectangular plan. The break from the curved or circular plan to a rectangular plan spurred the development and understanding of wall and multi-story systems. By 6500-6000 BC, two-story houses had been built in Anatolia and Cyprus, where the first story walls were made of mud brick with stone foundations and the upper story was typically timber-framed [6]. Houses in various locations in Greece (c. 4000 BC), had walls and roofs made of intertwined reeds fastened to a wooden framework, and plastered with clay, or were covered with brushwood, rushes, or straw with or without the impervious coating of clay [5]. Copper tools were already available at this time making possible a relatively easier way to cut wood across the grain and to fashion effective ways of joining timber members. Mortised joints, one of the most important techniques that made possible the construction of early buildings of various forms and sizes, probably developed near forested regions first and became more common and fully developed during the Iron Age. (It has been suggested that the builders of Stonehenge in England, c. 2000 to 1500 BC, had probably taken the mortise and tenon technique of the woodworker in fitting their huge trilithons together [7].)

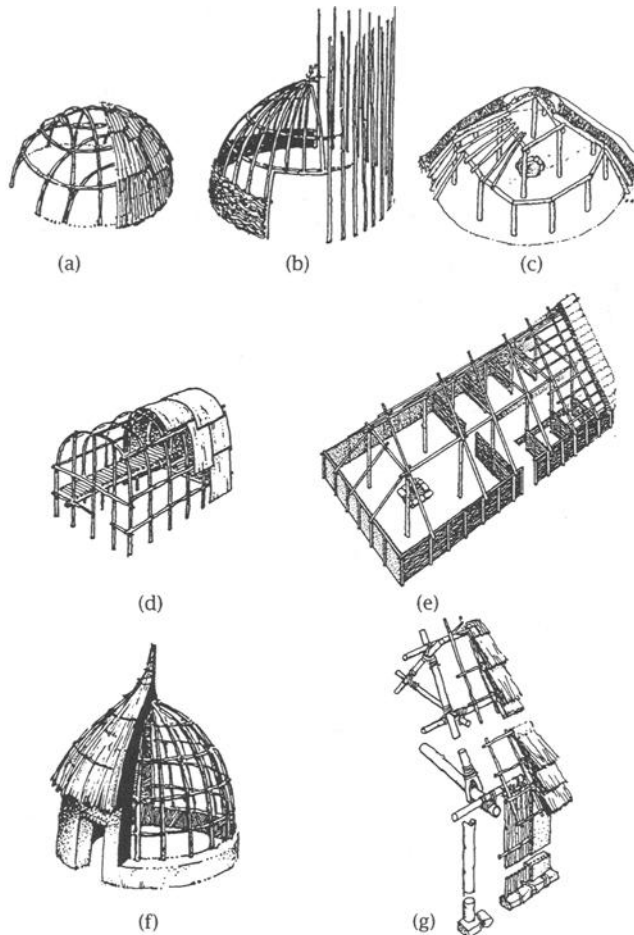


Figure 1 – *Types and forms of man's early shelter: (a) Palaeolithic round hut; (b) South African hut; (c) circular earth lodge; (d) lashed frame barrel vault with bark sheets; (e) lashed heavy pole frame with thatched roof; (f) hut with cylindrical clay wall and thatched roof; (g) lashed pole, thatched roof assembly [8]*

Slightly different forms developed later in other parts of the world. The Great Plains Indians of North America, for example, modified the hut to a cone of wooden poles lashed together only at the top (called a *tipi* or *tepee*). The taut membrane of animal skins pegged tightly about the frame held the structure firm by counterposing tensile stresses in the cover against the compressive strength of the poles [8]. In other places, the coverings were constructed of bark sandwiched between two sets of poles, one inside and one out [9]. Tartar tribesmen in Central Asia devised a demountable hut, called a *yurt* (Figure 2a), in which scissors trellis of pivoted wooden sticks carried a cone of poles socketed into a wooden ring. They lashed felted mats over the framed lattice to create a

warm and effective temporary shelter in severe and demanding climate. The yurt is light yet solid and strong and most important, the entire yurt could be broken down and folded together into pack load for ease of transportation [1, 8]. Wooden poles were extensively used in tents. The wooden framework of the mat tents of the Tuaregs of Africa consists of a grid of arches or semi-arches like a grid-shell (Figure 2b).

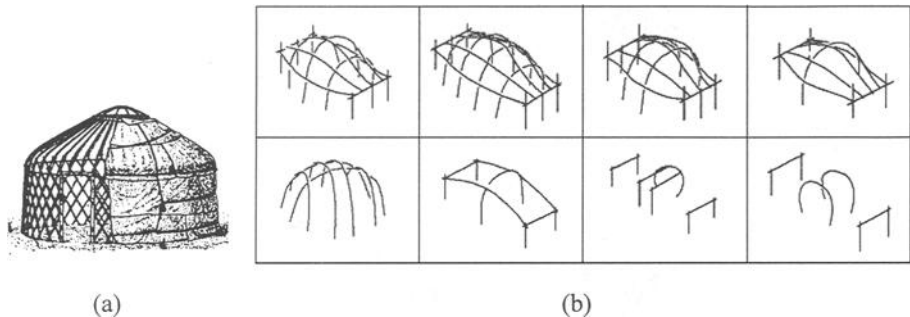


Figure 2 – *Wooden frame of pastoral shelters: (a) the yurt (from [10]); (b) the Tuaregs' mat tent [11]*

Requiring a different kind of building skills and sophistication, people in some other parts of the world lived in tree-dwellings and pile-dwellings. In the former, people built ladders and floors from tree branches and trunks. Pile-dwellings were common near lakes and swamps (also called lake-dwellings). Long wooden poles were driven securely into the bed of the lake or deep into swampy ground and wooden floors were built tying the poles together. Walls and roofs were typically thatched. The biggest collection of lake-dwellings in England, believed to have been built more than two thousand years ago in Somerset, showed woodworking and jointing techniques of “astonishing ingenuity” [12]. Once the floor was conceived as a wooden platform and raised above the ground, the concept of building on stilts was born [1]. This type of construction was, and still is, heavily practiced in parts of Asia, Africa, Polynesia and South America.

Ancient Civilizations

Mesopotamia and Egypt, the early centers of human civilization, and surrounding areas did not have a bountiful supply of timber. But the earliest deducible works of Egyptian architecture reflected wide use of supporting wooden frames hung with mats alongside buildings of brick and stone [13]. Recorded impressions from cylinder seals and mud plaster on the roofs of the Royal Tombs of the first dynasty (c. 3000 BC) suggest that the tent-palaces of the kings of Upper Egypt were timber-framed construction with vertical walls and a curved roof [13]. Both light wood-framed construction – reed hut with domed roof [5] – and wood frame construction made of laths covered with matting or frames with an infill of wattle and daub, clay mortar and rushes, or bricks [6, 13] were found in Egypt. Imported timbers from Lebanon and Syria were used in the construction of most buildings like temples and palaces [6], large vessels and fine furniture [1]. The first wooden furniture appeared in Egypt c. 3000 BC and furniture

with mortise and tenon joints were available c. 2600 BC. Leather thongs sometimes reinforced the joints. The early Egyptian woodworker had a high degree of craftsmanship. Furniture wood pieces were joined by dovetailing, by copper bands and, c. 1440 BC, by bronze pins or nails [14].

The famed Cedars of Lebanon framed Solomon's temple in the first millennium BC. Also, some timber bridges had been known to exist around this time, including an 11.0-m bridge at the palace of Minos, and Nebuchadnezzar's timber bridge which rested on more than a hundred piers to link the two parts of Babylon across the Euphrates river [7].

The earliest form of Greek temple, made up of a closed hall and an open portico bounded by posts that support the roof, was thought to be patterned after Greek dwelling houses, which were built with timber frames and infill of clay or sun-dried bricks [13]. Construction of the outer columns, including details of the abacus, eventually evolved from wooden pole into stone (e.g., the Mycenaean stone capital). Even after the transition from building in wood to building in stone occurred in Greek architecture, some temples continued to have timber roof supports, ceilings and/or wood panelling. Early Roman temples had similar components of wood. Later Roman monumental buildings were seldom made of wood, but dwelling houses and modest secular buildings in Pompeii, Herculaneum and Stabiae, for example, were typically timber-framed, with brickwork and mortar infill [13].

Development of Early Construction Forms

Wood was the dominant building material in Northern Europe during the Middle Ages and into the Renaissance. The carpenter was in the forefront of technical invention and wood played a predominant part in all operations of industry [4, 8]. Mumford [2] considered the time between the tenth to eighteenth century as the *eotechnic* or water- and-wood phase and viewed the developments in this period as the foundations of the industrial revolution.

Developments in timber construction did not occur successively. The forms and methods of construction in various places depended on the type of available timber, the conditions of use, and the culture and local traditions of the people. In this section, the developments in early common forms of timber construction are discussed. Some forms and techniques had clearly been passed on from one place to another (e.g., countries within Europe and from Europe to North America), and in most cases modified locally, while others developed in various places in parallel with or independent of other developments elsewhere.

Palisade or Stave Construction

Early wood buildings with solid walls had split trunks set vertically edge to edge in or on the ground (Figure 3a). In one excavation in moor settlements in Neolithic Germany, solid walls of vertical split trunks embedded in a wall trench and a beam with mortise holes lying in front of the open hearth were found [5]. Although this *palisade* or *stave* type of construction was easy to erect, the trunks were soon found to be susceptible to dry rot when embedded into the ground. Thus, horizontal sills of wood or masonry

were soon introduced. A horizontal plate at the top of the staves was later added and the timber trunks were grooved along the side and tongues of wood let in.

This stave construction continued through the Iron Age to Norman times (especially in building churches in Denmark, Norway, England and parts of central Europe) and was brought to North America by early French settlers along the Mississippi River [5, 15]. It is estimated that between eight hundred to a thousand stave churches, in Norway alone, were built from the eleventh to the fourteenth century [16]. More than thirty stave churches survive today (e.g., the Greensted church in Essex, England built c. AD 1013 [1]). The early staves were half-trunks but were trimmed to form planks, with flat faces front and back, or squared timbers in later buildings [5].

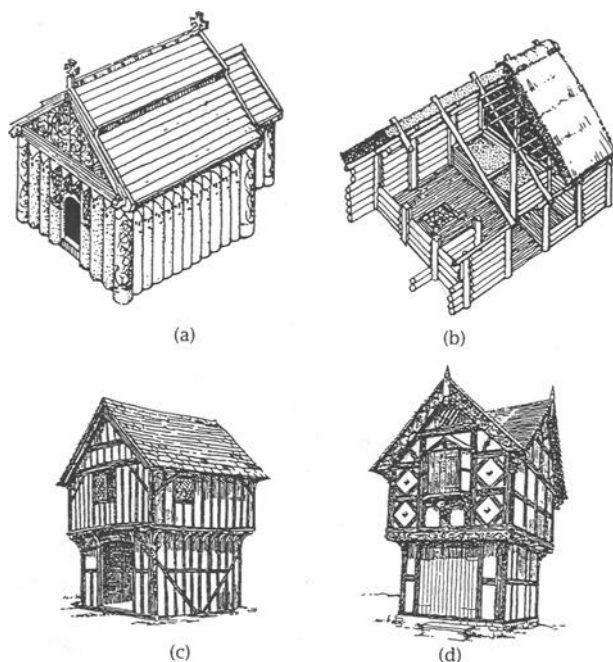


Figure 3 – Timber frame construction: (a) palisade or stave [8]; (b) log-cabin [8]; (c) straight-braced frame with post and pan construction [5]; (d) square-paneled frame construction [5]

Log-Cabin Construction

Large farmhouses in Central Europe c. 900 BC had logs placed horizontally on top of one another to form walls [5]. This type of construction came to be known as *log-cabin* or *log construction* (Figure 3b). There was a settlement of more than a hundred log houses in Poland c. 700 BC [1, 13]. Archaeological evidence and early historical accounts suggest the use of log construction in Scythia and Germany from the Neolithic period [1]. The first buildings were constructed of peeled logs only roughly notched at the corners [8], then the logs were later grooved on the underside and a variety of jointing

techniques at the building corners were developed. Square-hewn logs or sawn planks were also used later, especially after the establishment of saw mills at the beginning of the sixteenth century. This development allowed cutting of planks with mortises and tenons along their length and the development of better, weatherproof joints. Unlike walls with round logs which previously could be joined only at the corners, those with rectangular interlocking logs could be joined anywhere along the length of the wall allowing better transfer of loads. Larger structures, including five- and six-story log buildings, became feasible.

Many domestic, farm and church buildings in Europe, Russia, Asia Minor and the Himalayas were of the log-cabin type. This type of construction had also been practiced in Japan as early as the Yayoi Period (250 BC to AD 200) [17]. Eastern Europe has been called the “home of log-building” [1] because of its extensive use of this type of construction using the axe as the main tool. One log building that had been built as early as the thirteenth century and others built later still survive today. Some buildings had mixed log-cabin (usually the first story) and stave (usually the second story) construction. One of the most magnificent of all log buildings is the Church of Transfiguration in Kizhi, Russia, which was built in 1714 (Figure 4). It consists of a series of octagonal plans set on top of each other in pyramidal form, surmounted by a total of twenty-two onion-shaped domes which were clad with elongated lozenge-shape wooden shingles fixed in place by wooden pegs. The creative and masterful use of wood in this structure “represents the highest attainment in Russian wooden architecture” [1] and makes it “among the architectural masterpieces of the world” [13].



Figure 4 – *The Church of the Transfiguration in Kizhi, Karelia, Russia (photo by J. Loferski)*

The log-cabin type of construction was introduced in North America by Swedish (first along the Delaware River), Finnish, Scotch-Irish and German immigrants during the mid-seventeenth and early eighteenth centuries [1, 15]. This type of construction can be thought of as the forerunner of the modern shear structural system [18].

Timber-Frame Construction

Timber frame construction probably developed from early tent-like frame to the simple cruck frame, to cruck frame with collar and tie beams, and eventually to the post and truss/rafter frame [1], Figure 5a. In any case, timber frame or half-timber construction had been practiced as early as 6500 BC [6]. In some parts of Europe (c. 5500-2500 BC), dwellings of rectangular plan with single or multiple rows of posts to support the roof were common [5]. Buildings with elongated rectangular plans (e.g., Figure 1e), allowing the addition of as many bays as needed, were constructed because farmers needed larger buildings to store grain and crops. Frames were first made of lashed poles and the roofs were constructed of rafters and purlins and thatched (Figure 1g). The walls were interwoven branches or wattle or, in forest zones, split logs [5, 8]. Jointing methods evolved from lashing to notching to cutting of mortises and tenons. The latter made possible the construction of timber buildings of various forms and sizes.

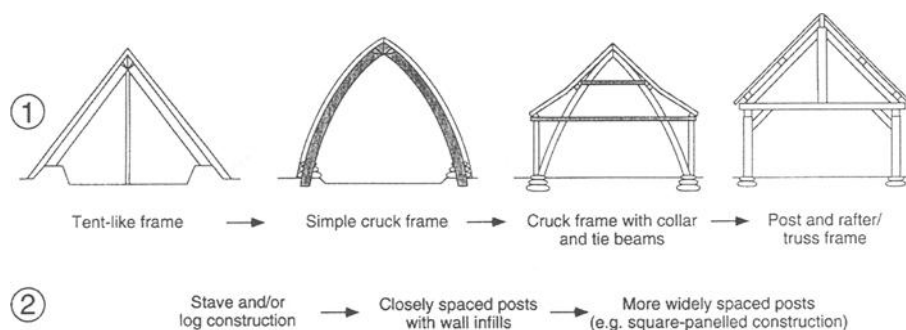


Figure 5 – Possible evolution of heavy timber construction: option (1) is from reference [1]; for option (2), see also Figure 3

In places where stave and/or log constructions were common, builders learned to dispense with the heavy log or plank infills between posts. The solid walls required a lot of timber and in areas with diminishing supply, the practice of filling interspaces between upright timbers with other materials such as plastered panels became more common. Many variations in the construction of these walls were practiced in the next fifteen hundred years [5] – from the simple to the more complex wattle and daub, to bricks laid in various patterns. The solid timber walls were replaced first by closely spaced posts (e.g., Figure 3c) – with continuous timber uprights from floor to ceiling, mortised top and bottom into horizontal wall-plates and sills – and later, as timber demand for other uses increased and the available supply for buildings decreased, by more widely spaced

posts (Figure 3d) – with horizontal timbers connecting the timber posts that divided the framework roughly into squares (known as *square-panelled* construction).

English immigrants brought these construction techniques to North America during the seventeenth and eighteenth centuries. The classical *saltbox* type of house in New England – with the exclusion of the brick and covering of oak or pine clapboards – evolved from adaptations to the pattern of the English timber frame house to provide more protection against the severe weather. There are about eighty seventeenth-century timber-frame buildings that are still extant in the United States (US) [1].

In Europe, the frames were universally made with oak timbers. The developed skills of the carpenters in the Middle Ages enabled them to construct some of the largest (six to seven stories in height) and most imaginative buildings of the Middle Ages and the Renaissance [1]. It is thought that timber framing (of the type discussed here) reached its highest development during the Renaissance, when glass became more available and used in windows, and decorative infill panels were common.

Roof Beams, Arches, and Trusses

A roof structure is a fundamental part of any building. Timber had a significant role from the earliest roofs ever built and has remained one of the most common material for roof-building. The simplest use of wood in roofing is as horizontal beams, e.g., in primitive pit dwellings or simple frames, or as poles or rafters leaning on a central post. Later combinations of these two basic types developed into other roof types that reached its potential in the *hammer-beam* roof, where short horizontal beams, or hammers, supported by wall-posts and braces project from the face of the wall to support the main arch (Figure 6). This type of construction allowed construction of buildings with spans as much as 20 m without any intermediate support. The roofs of the Westminster Hall, built in 1395, and the Eltham Palace, built in 1405 in England, are prime examples of this construction type. The hammer-beam roof of Westminster Hall was more of a stiffened arch than a real modern truss [19].

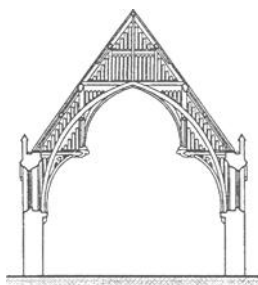


Figure 6 – *The hammerbeam roof structure (from [20])*

Unlike the concept of a modern truss, where all joints are assumed pinned, the traditional historic trusses consisted of flexural elements strengthened by a system of additional ties, struts, etc. [21]. Bracing of posts against overturning and rafters against

in-plane and lateral instability were probably one of the first ways braced timber systems developed [22]. When joints that resist tension forces were needed, timber members were interlocked together with more complicated mortises and pegged by wooden dowels. (It is relatively more difficult to make joints capable of transmitting and developing the tensile strength of the wood fibers.) Common rafters with braced collar and scissors, then trusses with queen posts, king post, and other combinations soon appeared. Once more effective tension joints were developed, especially with the increasing use of iron ties and dowels, longer span trusses were built. Gauld [23] provides a brief overview of European timber roof trusses prior to 1900.

Wooden deck bridges have been built using simple beams, strutted beams and arches since the Roman times. Timber arches were used in a bridge that was built by Emperor Trajan's troops over the Danube River in AD 106 [19, 24]. Through-truss timber bridges evolved from these simple forms and by the thirteenth century were common in Switzerland and Germany. Influenced by Swiss and German builders, Palladio built wooden arched-truss bridges, spanning more than 30 m, in Italy (c. 1550) (see also [22]). The Grubenmann brothers (Germany, c. 1778) built many long-span timber bridges – one had a maximum span of about 118 m [24, 25]).

It is noteworthy to point out that the first known glued laminated timber was used in Wiebeking's bridge in Altenmarkt, Bavaria in 1809 [26]. Two years earlier, Wiebeking also pioneered the use of mechanically bolted timber laminations in arch bridges; one of his bridges spanned 61 m. Similar bridge construction methods were practiced in France in the 1820's and in England in the 1830's.

Many timber truss bridges were built in Europe, Russia and the US in the beginning of railroad construction. James [25, 27] provides an extensive discussion of the evolution of timber bridge trusses to 1850 (see also *American Wooden Bridges* [28]).

Oriental Buildings

Far Eastern timber architecture developed independently of European efforts and was already fully developed by the time East-West contacts became more established. Timber construction in China dates back to the first millennium BC, and possibly earlier. A common method of connecting timber beams and columns was with the use of a bracket set – composed of a cap-block, a wooden block in the shape of an inverted truncated pyramid placed on top of the column, and two pieces of wood fitted into the cap-block perpendicular to each other (Figure 7a). The use of this jointing technique in Chinese architecture first appeared c. 1056-771 BC [29]. Its development reached maturity between the seventh and thirteenth centuries. The Yingxian Pagoda, with a 51.14-m high main timber body, was constructed in AD 1056 using extensive use of bracket sets and other traditional Chinese timber joinery. This structure has survived five major earthquakes and is still extant today. There are more than a dozen timber buildings in China that are at least one thousand years old [29].

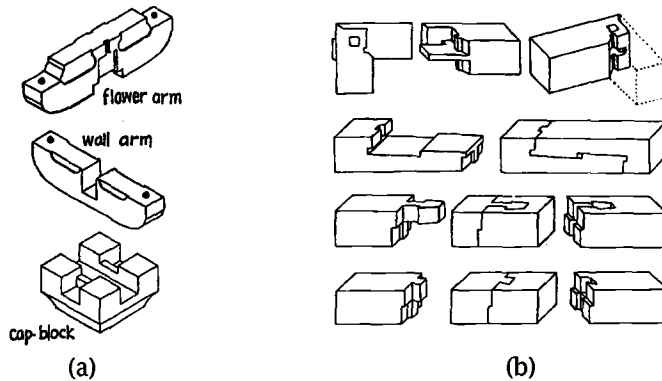


Figure 7 – Jointing techniques in oriental timber construction: (a) Chinese bracket set [29]; (b) Japanese joinery [17]

Although timber buildings had been built in Japan for dwellings, palaces, storehouses and Shinto shrines and the family carpentry guilds had already been formed long before Buddhism was introduced in the sixth century, the immigrant Buddhist carpenters from Korea brought Chinese building influence that contributed significantly to Japanese architecture and woodworking and jointing techniques. The Golden Hall and the five-story Pagoda of the Horyu-ji Temple (Figure 8a), built in the seventh century, are the oldest existing wooden buildings in the world [30]. The Golden Hall of the Todai-ji Temple (Figure 8b), built in the 8th century and rebuilt in the 18th century, is the world's largest non-engineered wooden building [30]. The Japanese mortise and tenon jointing methods developed into an art form and have become a unique feature of traditional Japanese architecture (Figure 7b). Many of these joints are still used today in the construction of Shinto shrines, Buddhist temples or residences; but no carpenter is deemed qualified to build all three types of construction since specific joints are typically reserved for only one type of construction [17].



Figure 8 – Japanese temples: (left) the Pagoda in Horyu-ji Temple and (right) the Todai-ji Temple

The basic structure of most Buddhist temples and pagodas is that of post-and-beam set on a base of stonework, sometimes with additional tiers of columns raised on top. Triangulation to support massive complex roofs were avoided in favor of the Chinese tradition of using short beams and/or brackets, supported by close columns transmitting loads vertically to the foundations. Unlike religious buildings in Europe, most Buddhist buildings have simple and symmetric plans (the most common shape being octagon, hexagon, circle or square), and uniform (with only gradual changes in height) and symmetric elevations. Some have simple rectangular plans with few subdivisions, with many-tiered tower structures standing in walled and colonnaded precincts [1].

Japanese domestic buildings, on the other hand, developed with little influence from the Chinese culture. The country's climate necessitated building on stilts to allow ventilation beneath the floor. The *shoin-zukuri* style, with free and asymmetrical plan, four- to five-inch square posts and widely overhanging roofs and verandas, has dominated Japanese domestic architecture since the eleventh century [1, 17].

The Industrial Revolution

Bricks were used extensively in the beginning of the industrial revolution in England and later in the rest of Europe. Wrought iron was first used mainly to strengthen timber and masonry systems. Wooden trusses, especially with forged straps and improved bolting, were used for roofs and road and railway bridges. During the late 18th century, cast iron started to replace heavy wooden beams and posts in multi-story (mill) buildings because of higher fire insurance premiums in the latter. By the 1850's, bigger and taller structures used wrought iron and steel. The former was used in the Eiffel Tower construction in Paris and the latter, together with the invention of the elevator, begat the skyscraper in New York and Chicago in the 1880's. The introduction of and developments in reinforced concrete in about the same time period further crowded the market for building materials. Wood, however, continued to be the chief material in furnishings of all types of buildings, including commercial, industrial and residential.

At the height of the industrial revolution in Europe, the US, with its abundant timber supply, was enjoying its wooden age which peaked at the turn of the 19th century, about two hundred years after it did in Europe [4]. American carpenter-builders had major contributions to timber construction. They are credited to have perfected the timber truss bridge in forms suited to mass production and capable of carrying railway loads [25, 27] and to have helped in the development and use of beam trusses, as opposed to European builders' arch trusses [22].

In 1833 in Chicago, the invention of the *balloon framing*, where uniform lengths of thin pieces of timber known as studs precisely spaced, braced and nailed together constituted the complete structural framework, started a revolutionary method of construction; it begat a later variation known as *platform framing*. This system of construction was made possible by the introduction of mechanical sawmills, making possible the supply of great quantities of planks and boards, and the mass production of nails. The balloon and platform framing methods sharply reduced the total labor cost of construction and made it possible to substitute relatively less skilled workers because there was no longer a need for mortise and tenon joints. The methods also lent itself to greater flexibility in construction options and to pre-fabrication. The flexible interior

arrangement of Victorian houses was thought to be partly due to the introduction of balloon-frame construction [15]. These light-frame methods of timber construction have since dominated low-rise residential and non-residential construction in the US.

Following the 1871 Chicago fire that destroyed a great part of the city, “what had been done in wood was re-created in iron” [14]. New constructions in cities followed the vogue of skyscrapers and of using steel and reinforced concrete. Developments in strength of materials and structural theory accelerated during this period. The modern period of structural engineering is thought to have started when Whipple published a method of truss analysis *c.* 1850 [31]. With the introduction of engineering analysis and calculations, knowledge of the mechanical properties of building materials became a necessity. Strength properties of common timber species started to get published in Europe, North America and Australia in the second half of the 19th century.

Twentieth Century Developments

Research

While strength testing of timber had been part of the early developments in strength of materials and theory of elasticity starting in Galileo's tests of timber beams *c.* 1638 [24, 32], systematic research on wood as a material started only at the turn of the twentieth century. Forest products studies were first conducted in a few universities and government institutions and in 1910, the US Forest Products laboratory (FPL), the world's first, and for sometime the only comprehensive laboratory dedicated to wood research, was established in Madison, Wisconsin [4]. Out of the wood mechanics studies that ensued emerged a system of grading rules, working stresses, and design specifications that formed the basis for the use of wood as a structural material. The move from a trial-and-error approach in the use of wood (as the woodworker did from the beginning) to one with a scientific basis was in high gear. Researchers studied the effects of fire, drying and preservatives on mechanical properties, and the design and behavior of connections and wood assembly systems, and published this wealth of knowledge in the first edition of the *Wood Handbook* in 1935. Work at FPL contributed in the preparation of wood-related standards (e.g. ASTM) and codes, provided practical information to engineers, builders and homeowners, assisted the forest products industry and influenced forest products research worldwide [33]. Similar application of research results to practical timber construction issues also happened in Europe and Australia [34].

Recent developments in wood engineering and construction, including the state of the art and research needs at the end of the twentieth century, are reviewed in [35].

Key Products

Many wood and wood-based products have been introduced in the twentieth century, but two stand out in their unique contributions to wood construction: plywood and glued-laminated (glulam) timber.

With animal and vegetable glues, the plywood that was produced early this century was restricted for interior uses, e.g. cabinets, doors and furniture. Production did not expand significantly until after World War I. The use of synthetic resin adhesives in 1935

in bonding veneers produced “exterior” plywoods that were used in boats and outdoor construction [15]. The use of resin-bonded Douglas-fir plywood grew steadily and in 1963 the first Southern pine plywood was introduced. These panels were used as underlayments for floors, walls and roofs and/or as finish materials. Plywood's main structural use is to transfer in-plane shear forces, as in wood shear walls and diaphragms, and shell-type structures. Since then, other wood-based panels, e.g. oriented strand board (OSB), have been introduced for similar applications.

Beams and arches have been built up from layers of timber for many hundreds of years, and the first glulam had been used in a 42-m span bridge in 1809 in Bavaria [26]. In Australia, glulam members have been found in a church building, which was built in 1868 in Victoria [36]. The German carpenter Otto Hetzer obtained the first known patent on glulam construction in 1900 and pioneered its use early this century [37]. However, it was not until the 1930's when wood adhesives that produced bonds that are stronger than the bond of the wood cells became available. The potential of glulam's extensive application in construction began to be realized. The practical length and cross-sectional dimensions of structural wood members increased by several factors and curved and/or tapered members became possible. With greater design flexibility, dimensional stability and strength of glulam members, new uses for wood became possible.

Among the most striking glulam structures in the US are the TRESTLE aircraft and communication system test facility in New Mexico (the largest glulam structure in the world with nearly 15,350 cubic meters of glulam) and the Tacoma Dome in Washington state (with a clear span of 162 m). Rhude [38] has presented the history of glulam construction in the US and Medwadowski [18] has discussed the developments of modern wood spatial structures in the US and presented several constructed systems.

Modern developments in practical timber construction have been realised because of developments in jointing methods. For example, the introduction of metal connector plates in the US in the 1950's, and in other parts of the world in the 1960's, launched the automated truss industry and significantly contributed to the viability of timber construction for residential and low-rise buildings. Modern spatial timber structures rely heavily on specialised connectors to transfer loads. Connections between and to wood components have been both a catalyst and a barrier for timber use in construction [39].

Residential Buildings

Over ninety percent of residential buildings in North America are of light wood-frame construction (i.e., platform frame or 2x4 construction). In the US, sixty percent of the available sawn wood is used structurally. Other residential building uses of wood include partitions, door and window frames, etc. In California alone, it has been roughly estimated that the total value of wood in residential housing is at least US\$100 billion in 1994 [40]. In Japan, eighty percent of the available sawn wood is used structurally (mostly in post-and-beam construction as pointed out earlier). In Australia, about seventy five percent of single detached housing is of light timber frame construction.

Common timber products used in residential buildings in North America include sawn timber, plywood, particleboard, OSB, laminated veneer lumber (LVL), trusses, I-joists, glulam and parallel strand lumber (PSL). Similar products are used in non-residential building construction.

Non-Residential Buildings

Commercial and industrial timber buildings typically have portal or post-and-beam frames made of glulam, PSL and/or LVL. Truss-beams and I-joists are sometimes used in floors and roofs. In North America, platform frame low-rise commercial buildings are not uncommon, sometimes walls on 2x4 frames are built atop a first story reinforced concrete frame. Tilt-up warehouse buildings typically have glulam and plywood or OSB roof frames. Expanding the structural timber use in the non-residential market has long been a priority in North America.

Large-span timber structures, most of which use glulam, had been built all over the world [41, 42]. The three 1994 Winter Olympics stadiums in Norway were roofed with glulam trusses. The Hamar Olympic Stadium, has the form of an inverted Viking ship and was roofed with arched glulam trusses with spans varying between 30.0 m to 96.4 m. The waste recycling hall in Vienna, with a cone saddle shape (170-m diameter at the base), has a shell-rib glulam structural framework where each radial rib (85-m span) is suspended from centrally located reinforced concrete mast. Natterer et al. [41] provided more details of this and one hundred and sixty five other types of timber roofing structure that have been constructed for large assembly halls and multi-purpose areas. Sample schematics of selected projects are shown in Figure 9a.

The timber grid-shell structure developed by Otto in Europe and employed in a 61-m span exhibition hall in Mannheim in 1975 was a major development in lightweight timber roof construction. A very similar system was used in building two temporary pavilions in Japan for the 1988 Nara Silk Road Exposition [30]. The lattice framework was composed of two-fold sawn 7-cm x 4-cm lattice members that were connected by a 9-mm bolt. Following a similar concept, a 25-m x 25-m timber grid-shell exposition hall (Figure 9a, bottom drawing) has been constructed in Switzerland [43].

Sedlak [42] and associates prepared a computer database of worldwide projects using modern timber spatial systems and studied the statistics and trends of building shape, structure and architectural applications of more than a hundred of these buildings – with types ranging from shells, space-grid, lightweight trusses, arches to suspended structures. They found that sixty four of the buildings he considered were high quality permanent buildings for sport, assembly, multi-purpose, commercial and dwelling uses. Twenty nine percent were quality temporary applications such as structures for exhibition, environmental protection, experimental and open air stages. The rest were used for lower cost industrial buildings of medium duration. Considering the building volume shape, domes are predominant in multipurpose and industrial, exhibition, assembly and indoor sports buildings. Other building forms include (in descending order of occurrence) prisms (mainly saddle/tension prisms), vaults (mainly barrel vaults), cones, pyramids and cylinders. Considering structural type, grid-shells (compression and suspension types) are the most dominant across the entire range of applications. Membrane-shell and ribbed shells are tied in second. Cladding is integrated with the structure and is often either a stressed roof membrane or a conventional waterproof membrane on timber boarding. Timber arches, folded structures, spatial grid, cable-net-shell, truss and spatial frame have also been used. A new field of development is suspension timber shells, which are able to access a much wider range of shapes.

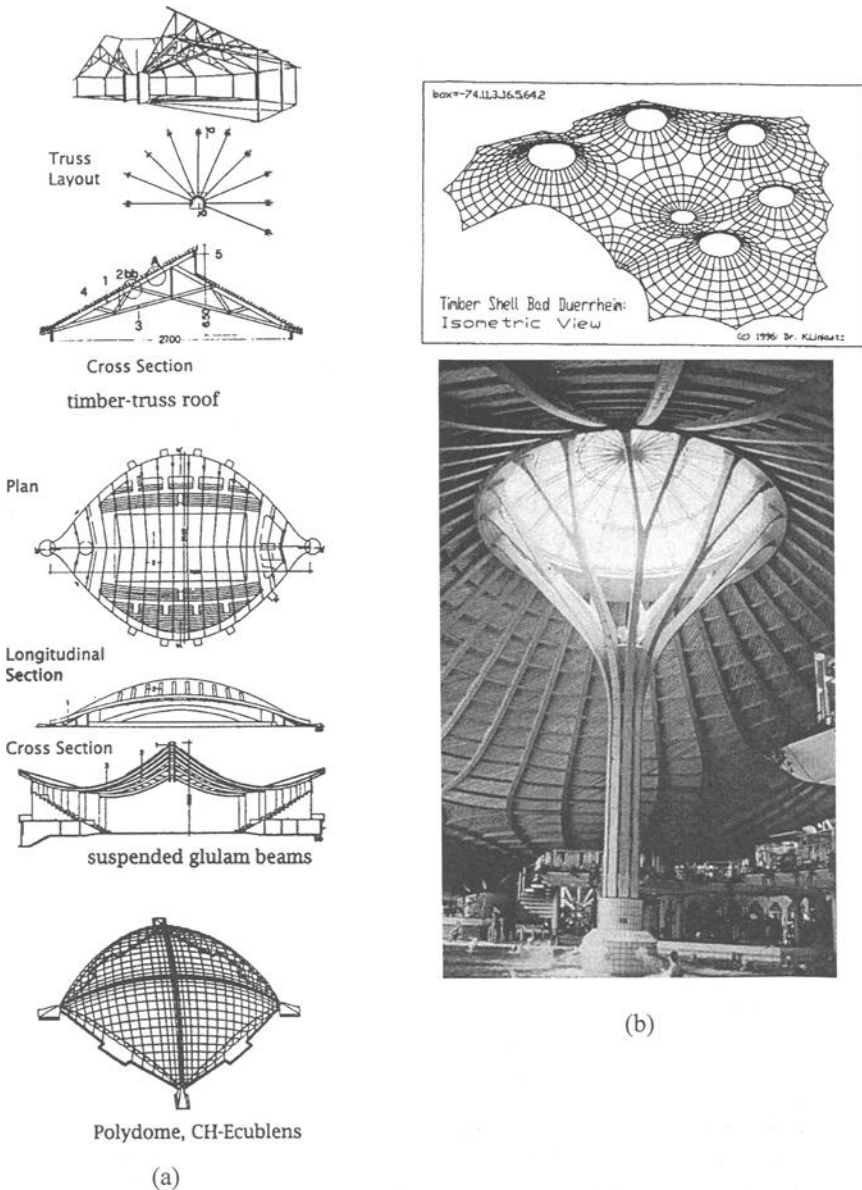


Figure 9 – Modern timber structures: (a) Polydome from [43], other drawings from [41]; (b) Health spa in Bad Durrheim, Germany: (top) isometric diagram of the structure [44] and (bottom) interior column support

State-of-the-art techniques in integrated formfinding and numerically controlled premanufacturing have been used in the design and construction of glulam timber shells (e.g., the health spa in Bad Duerrehim, Germany; Figure 9b). Timber structures are not limited to double curved surfaces, which are typical for membranes and pre-stressed cable nets. When properly designed and manufactured, timber load bearing elements can be shaped almost arbitrarily with respect to curvature and torsion as demonstrated in the free-form timber shell roof of the Hoelderlin-Haus in Maulbronn, Germany [44]. In this project, up to two thousand small timber pieces were glued together to form glulam sections that accommodate the rapid differential change in the longitudinal development of the beams.

Past Contributions and Future Outlook

The versatility and usefulness of wood in its many forms and adaptations throughout man's existence have made it a major contributor to the advance of civilization and the development of society, generally, and to developments in architecture, structural forms and building construction, specifically. Wood played an important part, as test specimens and/or test frames, in the early developments of the field of strength of materials [24, 32]. Elements of ancient Greek architecture evolved from timber prototypes [13], the Medieval Gothic vault evolved from simple timber roofs and clay dome or shell vaults with wooden armature as a reinforcing mesh [8], modern structural framing (in steel and reinforced concrete) and truss systems were first modeled from timber framing and truss system concepts, and jointing techniques and details for iron and steel were initially patterned after those in timber [19]. Wood and wood-based products have been used extensively as formworks and/or falseworks in almost all reinforced concrete structures. Public buildings and residential buildings made of timber have provided man shelter throughout history.

Recent advances in structural testing and analysis, manufacturing and construction techniques, computer technology, materials science and timber engineering technology, and developments in standards and building codes can be harnessed altogether to continue the long and rich tradition of building with wood into the future. There seems to be a renewed enthusiasm about the aesthetic properties of timber [21, 30, 45], a great interest in the use of lightweight materials for space structures [46] and a bright potential in the use of timber in integrated formfinding and premanufacturing of shell structures [44].

Wood, our only basic renewable resource, has been an important part of man's built environment and – if we properly manage our forests, recognize the changing natures of timber supply and building demand, continue to advance the state-of-the-art in our knowledge of the properties of old and new wood and wood-based building products and their fastening systems, and in the analysis, design and construction techniques, and apply them in practice – may continue to be so in the future.

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